Recommendation Rabbit Hole Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	**Time**	30-40 min
Difficulty	Medium	**ID**	recommendation-rabbit-hole

Big idea: A feed can become a path dependency: early clicks shape later knowledge possibilities.

Overview

Students simulate repeated recommendations and observe how small choices can narrow a user's knowledge environment.

Student Prompt

Inspect Recommendation Rabbit Hole Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Recommendation Rabbit Hole Stimulus A	Student-facing stimulus 1: Starting interest: exam stress; feed branches into study tips, productivity hacks, anxiety posts, tutoring adverts, and sleep science. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Recommendation Rabbit Hole Stimulus B	Student-facing stimulus 2: Starting interest: climate action; feed branches into science explainers, activism, conspiracy claims, policy debate, and lifestyle content. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Recommendation Rabbit Hole Reveal / Twist	Student-facing stimulus 3: Starting interest: climate action; feed branches into science explainers, activism, conspiracy claims, policy debate, and lifestyle content. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — A feed can become a path dependency: early clicks shape later knowledge possibilities.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Recommendation Rabbit Hole Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Recommendation Rabbit Hole Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.

#	Question for students	Teacher key / cue
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	When does personalization become distortion?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how a feed can become a path dependency: early clicks shape later knowledge possibilities.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

• Prepare enough cards that each round can narrow without becoming repetitive.

• Keep content fictional and school-safe while preserving emotional and credibility differences.

Ready-to-Use Examples

• Starting interest: exam stress; feed branches into study tips, productivity hacks, anxiety posts, tutoring adverts, and sleep science.

• Starting interest: climate action; feed branches into science explainers, activism, conspiracy claims, policy debate, and lifestyle content.

Suggested Timing

Stage	Teacher move	Watch for
1	Give each student or group a starting interest card.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students choose one item from a small feed of mixed content.	Assumptions, confidence shifts, evidence claims, and language choices.
3	The algorithm adds more cards similar to the chosen card in the next round.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Repeat for three or four rounds and map how feeds diverge.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief what users might come to believe is normal, popular, or true.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

• The user keeps choosing, but the menu of choices keeps changing.

• At what point does personalization become a knowledge environment?

Common Misconceptions

• Students may treat a tool's output as neutral because it feels automatic.

• Students may focus only on user error and miss design incentives or hidden rules.

Assessment Look-Fors

• Identifies the rule, design choice, or incentive shaping knowledge.

• Distinguishes access to information from justified knowledge.

Differentiation and Adaptations

• Support: run two rounds only.

• Challenge: add engagement feedback so popular cards become more likely next round.

Extension Tasks

• Students design an alternative feed rule that balances curiosity, reliability, and wellbeing.

• Connect the path map to a TOK exhibition object such as a phone screen.

Related Activities

• Algorithmic Feed Simulation

• Search Engine Ranking Game

• Deepfake Detection Challenge